Convertir Word a PDF

Executive Technical Feature & Client Deliverables Report

Next.js 14 • TypeScript • Production Ready

Date: November 2026

Target Market: Spanish-speaking users

Project Summary: A high-performance, responsive online utility designed for converting Microsoft Word documents (.DOC and .DOCX) into vector-crisp PDF files. Features an authentic backend conversion pipeline, batch multi-file processing, instant in-app multi-page preview, custom filename editing, page management (rotate/reorder/delete), real compression, AES-256 password protection, and automated 1-hour secure file cleanup.

1. Complete Feature Specifications

1. Real Conversion Engine

Backend

Converts DOCX and DOC files with full preservation of typography, headings, tables, bullet points, headers/footers, and embedded images into true vector PDFs.

2. Filename Customization

UX / Download

Inline editable filename input. Strips illegal symbols, handles accents/Unicode, prevents double extensions (.pdf.pdf), and delivers the custom name on download.

3. Multiple File Upload

Multi-File

Supports drag-and-drop and multi-file picking. Each file is validated independently; an invalid file displays an individual badge without breaking the rest.

4. Batch Conversion Queue

Performance

Processes multiple documents with concurrency control (max 2 parallel workers) to prevent server overload. Real-time status: Waiting, Processing, Completed, Failed.

5. Download All as ZIP

Packaging

One-click batch download button. Packages all successfully converted PDFs into a clean, de-duplicated standard DEFLATE ZIP archive using custom filenames.

6. In-App PDF Preview

Interactive

Multi-page preview modal with Next/Previous navigation, page counter (Page X of Y), Zoom In (+), Zoom Out (-), and Fit-to-width without downloading.

7. PDF Page Settings

Layout

Configurable before conversion: Paper Size (A4 standard or Letter) and Orientation (Portrait or Landscape) applied directly to the print stylesheet.

8. Page Organizer (Rotate & Reorder)

Post-Process

Interactive modal to rotate individual pages (90° clockwise), reorder sequence (move left/right), and delete pages with instant undo. Exports a separate modified PDF.

9. Real PDF Compression

Optimization

Three levels: Low, Medium, High. Compresses content streams and duplicate objects. Reports true byte savings and percentage reduction honestly without fake stats.

10. Password Protection

Security

Encrypts PDF with AES-256 standard encryption. Requires confirmation, validates length, and verifies decryption in memory before offering download.

11. Enterprise Privacy & Security

Compliance

Server-side magic byte inspection (PK\x03\x04 and \xD0\xCF\x11\xE0). Isolated UUID temporary paths. Passwords never logged. Automated 1-hour deletion cycle.

12. Mobile-First Spanish UI

SaaS Design

100% natural Spanish copy tailored for Spanish-speaking users. Clean SaaS layout inspired by top tools (Smallpdf, iLovePDF). Zero horizontal overflow on 360px+ screens.

2. Backend API Architecture

API Endpoint	HTTP Method	Purpose & Responsibilities
`/api/convert`	POST	Converts DOC/DOCX to vector PDF with custom filename, paper size (A4/Letter), and orientation.
`/api/download`	GET	Serves PDF as direct download (attachment) or browser viewer (inline=true) with dynamic custom filename.
`/api/batch-zip`	POST	Packages selected converted files into a standard ZIP archive with filename de-duplication.
`/api/compress`	POST	Optimizes PDF streams and deduplicates objects across Low, Medium, and High compression levels.
`/api/protect`	POST	Encrypts PDF using AES-256 password protection and verifies decryption capability in memory.
`/api/pages`	POST	Reorders, rotates (90° steps), and removes selected pages using pdf-lib; exports modified PDF.

3. Quality Assurance & Verification

All 6 core modules and endpoints have been thoroughly tested and validated against real Word documents (.docx with tables, formatting, and images). Production compilation (`npm run build`) completed with 0 errors and 0 type warnings.